Chapter 4 Study Sheet

Transformations – everything you need on one page.

Transformation	Rule	Rigid motion?
Translation along ⟨a, b⟩	(x, y) → (x + a, y + b)	Yes
Reflection in the x-axis	(a, b) → (a, −b)	Yes
Reflection in the y-axis	(a, b) → (−a, b)	Yes
Reflection in y = x	(a, b) → (b, a)	Yes
Reflection in y = −x	(a, b) → (−b, −a)	Yes
Reflection in x = k	(a, b) → (2k − a, b)	Yes
Reflection in y = k	(a, b) → (a, 2k − b)	Yes
Rotation 90° about O	(a, b) → (−b, a)	Yes
Rotation 180° about O	(a, b) → (−a, −b)	Yes
Rotation 270° about O	(a, b) → (b, −a)	Yes

The Three Rigid Motions

- Translation (4.1): slides every point the same distance in the same direction, along a vector ⟨a, b⟩.
- Reflection (4.2): flips a figure over a line. The line of reflection is the perpendicular bisector of every segment joining a point to its image.
- Rotation (4.3): turns a figure about a fixed center. All rotations are counterclockwise unless you are told otherwise.
- All three preserve length and angle measure, so in every case the image is CONGRUENT to the preimage.

Working Backward and Composing

- To find a PREIMAGE from an image, undo the rule: subtract what you added, and reverse the switch or the sign change.
- Two translations in a row: just add the vectors. ⟨a, b⟩ then ⟨c, d⟩ is the single rule (x, y) → (x + a + c, y + b + d).
- A glide reflection is a translation followed by a reflection in a line PARALLEL to the translation vector.
- Order usually matters. Do each step in the order it is written, and label the first image with ′ and the second with ″.

Symmetry

- Line symmetry: a reflection in some line maps the figure onto itself. That line is a line of symmetry.
- Rotational symmetry: a rotation of 180° or less about the center maps the figure onto itself.
- A regular n-gon has n lines of symmetry and rotational symmetry every 360°/n degrees.
- Careful: a rectangle that is not a square has 2 lines of symmetry, NOT 4 — its diagonals are not lines of symmetry.
- A parallelogram that is not a rectangle or rhombus has 0 lines of symmetry but DOES have 180° rotational symmetry.

Congruence Transformations (4.4)

A congruence transformation is a rigid motion, or a composition of rigid motions. Two figures are congruent if and only if such a transformation maps one onto the other.

Composition Theorem: the composition of any number of rigid motions is itself a rigid motion — so the final image is still congruent to the original.

Reflections in two PARALLEL lines = a translation, perpendicular to the lines, of TWICE the distance between them. (Lines 3 apart give a translation of 6.)

Reflections in two INTERSECTING lines = a rotation about the point where they meet, through TWICE the angle between them. (Lines at 45° give a 90° rotation.)

Dilations (4.5)

Centered at the origin, scale factor k: $(x, y) \to (kx, ky)$. About a center C(a, b): $(x, y) \to (k(x - a) + a , k(y - b) + b)$.

k = image length ÷ preimage length. $|k| > 1$ enlarges; $0 < |k| < 1$ reduces; a negative k also puts the image on the opposite side of the center (a 180° turn as well).

A dilation preserves ANGLE measure but NOT length, so it is NOT a rigid motion unless $|k| = 1$. The image is SIMILAR to the preimage, not congruent.

Similarity Transformations (4.6)

A similarity transformation is a dilation, or a dilation together with rigid motions. Two figures are similar if and only if such a transformation maps one onto the other.

In similar figures corresponding angles are CONGRUENT and corresponding sides are PROPORTIONAL — every ratio equals the scale factor.

Every congruence transformation is a similarity transformation with k = 1. The reverse is false: similar figures are congruent only when k = 1.

Careful: a dilation does not "tilt" anything. Angles are unchanged, which is exactly why the shape survives and only the size moves.